Experiment 3: Torsion of a circular shaft

Objectives

Understand the behavior of a circular shaft subjected to a twisting moment and test the predictions of the theory of torsion in circular shafts.

Background

Shafts are slender members that resist external loads primarily by way of twisting. Shafts are used in the transfer of mechanical power from one point to another. They may also be used to provide stiffness to a sprint with respect to rotation.

The theory of twisting of circular shafts is used to compute the angle of twist, shear strain, and consequently stresses in a shaft subjected to a twisting moment. In this experiment, we will put this theory to test by comparing the experimental outcome and the theoretical predictions.

Before conducting the experiment, we must first answer the following questions:

1.) What are the theoretical assumptions made in the twisting of a circular shaft?
2.) For a circular shaft with shear modulus G and radius r, loaded at the ends by twisting moment M_t, show that the angle of twist ϕ at a distance z from the fixed end is governed by the equation:

$$\phi = \int_0^x \frac{M_t}{GJ} dz$$

where J is the polar moment of area of the shaft cross-section.

3.) Does torsion produce homogeneous deformation?
4.) What is the polar moment of area?
5.) How to calculate modulus of rigidity from a torsion test?

Some online resources

You may watch the following resources to refresh your memory on torsion of circular shafts.

Torsion of circular shaft (22:14 minutes)

Torsion test (2 mins)

[Torsion testing of Al sample to failure] (21 mins to failure)

[Torsion testing - II]

The experiment

Task 1:

The apparatus to load a circular shaft and measure the angle of twist is available in the ME lab. Your job is to set up shafts made of 3 different materials with 3 different values of twisting moment applied. The value of the twisting moment must be chosen such that the rotation at the end of the shaft remains small. For each value of the applied twisting moment, measure the angle of twist at end of the shaft. Take help from the lab staff and TAs if you need to. Compare these deflections with the theoretical formula of the angle of twist given in the "Background" section above.

Task 2:

Write a report on the entire exercise. The report must not exceed 3 pages. You can tailor your report as per the following proposed structure, but you are free to deviate from it if you wish to.

1.) INTRODUCTION: State your objectives, and the answers to questions 1-3 stated in the "Background" section above.
2.) PROCEDURE: Describe the details of the procedure used to conduct the experiment.
3.) OBSERVATIONS: Report all the experimental data recorded. Describe your observations, insights, and any calculations you perform. Use plots and tables to compare experiments with theory. Does your experimental data agree with the theoretical predictions? If not, explain the possible reasons.
4.) CONCLUSIONS: Describe what you understood from this entire exercise.